

Blanc's theorem

Th (Blanc): Soit G un groupe de Lie nilpotent simplement connexe et $\pi: G \times V \rightarrow V$ une représentation unitaire irréductible non triviale. Alors $H_d^*(G, V) = 0$.

Ingredients

1) Prop 1: $G = \text{top group second countable}$, $\pi: G \times V \rightarrow V$ a continuous representation of G on a Hausdorff top vector space. Suppose $\exists g \in Z(G)$ st $\pi(g) - \text{id}$ is invertible. Then $H_d^*(G, V) = 0$.

Notations: $\partial_k: \mathbb{C}G^{k+1} \rightarrow \mathbb{C}G^k$, $\partial_k(x_0, \dots, x_k) = \sum_{i=0}^k (-1)^i (x_0, \dots, \widehat{x_i}, \dots, x_k)$
so that $d_k: C(G^{k+1}, V) \rightarrow C(G^k, V)$ is $(d_k f)(\sigma) = f(\partial_{k+1} \sigma)$.

For $g \in G$, let $\tau_g: \mathbb{C}G^{k+1} \rightarrow \mathbb{C}G^{k+1}$, $\tau_g(x_0, \dots, x_k) = (x_0 g, \dots, x_k g)$

and $t_g: C(G^{k+1}, V) \rightarrow C(G^{k+1}, V)$, $(t_g f)(\sigma) = f(\tau_g \sigma)$.

Then $\tau_g \circ \partial = \partial \circ \tau_g$, $t_g \circ d = d \circ t_g$ and t_g commutes with the G -action on $C(G^{k+1}, V)$.

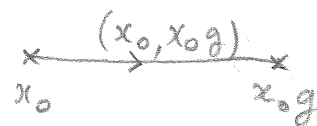
Therefore t_g induces $t_g^*: H^*(G, V) \rightarrow$.

Lemma: $t_g^* = \text{id}$.

Pf: Define an homotopy $\mathcal{K}_k: \mathbb{C}G^{k+1} \rightarrow \mathbb{C}G^{k+2}$
with $\mathcal{K}_{k+1} \circ \partial + \partial \circ \mathcal{K}_k = \tau_g - \text{id}$, such that

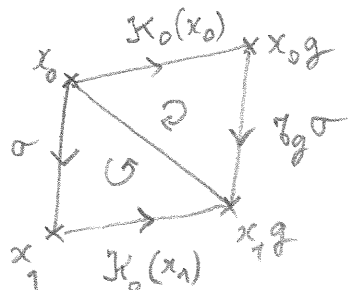
$h_k: C(G^{k+1}, V) \rightarrow C(G^k, V)$, $(h_k f)(\sigma) := f(\mathcal{K}_{k+1} \sigma)$,
is a G -equivariant homotopy: $h_{k+1} \circ d + d \circ h_k = t_g - \text{id}$.

* $\mathcal{K}_0 : \mathbb{C}G \rightarrow \mathbb{C}G^2$ with $\partial \mathcal{K}_0(x_0) = x_0 g - x_0$
 $\mathcal{K}(x_0) = (x_0, x_0 g)$ works.



* $\mathcal{K}_1 : \mathbb{C}G^2 \rightarrow \mathbb{C}G^3$ with

$$\mathcal{K}_0(\partial \sigma) + \partial \mathcal{K}_1(\sigma) = \tau_g \sigma - \sigma \quad \forall \sigma = (x_0 x_1) \in G^2$$



$$\mathcal{K}(x_0 x_1) = (x_0, x_0 g, x_1 g) - (x_0, x_1, x_1 g) \text{ works}$$

* In general $\mathcal{K}_k : \mathbb{C}G^{k+1} \rightarrow \mathbb{C}G^{k+2}$,

$$\mathcal{K}_k = \sum_{i=0}^k (-1)^i \mathcal{K}_k^i, \text{ with}$$

$$\mathcal{K}_k^i(x_0, \dots, x_k) = (x_0, x_1, \dots, x_i, x_i g, x_{i+1} g, \dots, x_k g) \text{ works}$$

(See Guichardet prop 7.1). $\square$

Pf of Prop 1: Let $g \in Z(G)$. One has for

$$\text{every } f \in C(G^k, V)^G : (t_g f)(x_0, \dots, x_k) = f(x_0 g, \dots, x_k g) =$$

$$= f(g x_0, \dots, g x_k) = \pi(g) \cdot f(x_0, \dots, x_k). \text{ Therefore } t_g f = \pi(g) f.$$

Suppose $\pi(g) - \text{id} : V \rightarrow V$ is invertible. Let $[f] \in H_d^k(G, V)$.

$$\text{One has } [(\pi(g) - \text{id})f] = [\pi(g)f - f] = [t_g f - f] = 0 \text{ by}$$

The previous proposition. Therefore there exists $c \in C(G^k, V)^G$ st

$$(\pi(g) - \text{id})f = dc; \text{ and } f = (\pi(g) - \text{id})^{-1} dc = d(\pi(g) - \text{id})^{-1} c.$$

Thus $[f] = 0$. $\square$

2) Hochschild-Serre spectral sequence.

Let $N \triangleleft G$ be a closed normal subgroup and $\pi: G \times V \rightarrow V$ a continuous representation on a Hausdorff loc. convex vector space.

$G \curvearrowright C(N^{k+1}, V)$ via $(g \cdot f)(x_0, \dots, x_k) = \pi(g) f(\bar{g}^1 x_0 g, \dots, \bar{g}^1 x_k g)$.

This induces $G \curvearrowright H_{\text{ct}}^*(N, V)$.

Moreover, when $g \in N$ and $f \in C(N^{k+1}, V)^N$, one has $(g \cdot f)(x_0, \dots, x_k) = f(x_0 g, \dots, x_k g)$. Thus, by Prop 1, N acts trivially on $H_{\text{ct}}^*(N, V)$ and so $G/N \curvearrowright H_{\text{ct}}^*(N, V)$.

The Hochschild-Serre spectral sequence relates $H_{\text{ct}}^*(G, V)$ to $H_{\text{ct}}^*(G/N, H_{\text{ct}}^*(N, V))$ under some additional hypothesis. As a special case one has:

Prop 2: Suppose G is a Lie group, $N \triangleleft G$ a closed subgroup. Assume V is Frechet and $H_{\text{ct}}^*(N, V)$ Hausdorff. If $H_{\text{ct}}^p(G/N, H_{\text{ct}}^q(N, V)) = 0$ $\forall p, q$ with $p+q=k$, then $H_{\text{ct}}^k(G, V) = 0$.

Dem. du thm : G nilpotent simply connected
 $\Rightarrow Z(G)^0 \cong \mathbb{R}^l$ with $l \geq 1$. Let $N < Z(G)^0$,
 $N \cong \mathbb{R}$. One has $\pi|_N(x) = \int_{\mathbb{R}} e^{itx} dE(t)$,

where $E =$ measure with projection values, $E: \mathcal{B}(\mathbb{R}) \rightarrow \text{Proj}(V)$.

Since $N < Z(G)$, for every borel set $B \subset \mathbb{R}$, the subspace $\text{Im}[E(B)] \subset V$ is G -invariant. Since π is irreducible

$$\exists t_0 \in \mathbb{R} \text{ s.t. } \pi|_N(x) = e^{it_0 x} \text{id}_V.$$

Two cases : 1) $t_0 \neq 0$. Then $\exists x \in N$ st $\pi(x) - \text{id}$ is invertible. Prop 1 $\Rightarrow H_{\text{ct}}^*(G, V) = 0$.

2) $t_0 = 0$. Then N acts trivially on V , thus π induces $\pi: G/N \times V \rightarrow V$, irreducible and non trivial.

$$\text{One has } H_{\text{ct}}^k(N, V) = H_{\text{ct}}^k(\mathbb{R}, V) = \begin{cases} V^{\mathbb{R}} = V & \text{if } k=0 \\ V & \text{if } k=1 \\ 0 & \text{otherwise,} \end{cases}$$

since $H_{\text{ct}}^1(\mathbb{R}, V) = \{ \text{affine action of } \mathbb{R} \text{ on } V \text{ with trivial vectorial part, without fixed points} \}$.

Thus $H_{\text{ct}}^p(G/N, H_{\text{ct}}^q(N, V)) = 0$ by induction on $\dim G$.

Now Prop 2 $\Rightarrow H_{\text{ct}}^*(G, V) = 0$. $\square$

Ref: P. Blanc, Sur la cohomologie continue des groupes localement compacts, Annales Scientifiques ENS, 1979.
A. Guichardet, Cohomologie des groupes topologiques et des algèbres de Lie, CEDIC, 1980.